

AI Quant Trading Platform - Software Requirements & Technical Design Document (Version 1.0)

1. Project Overview

Develop an AI-powered cryptocurrency trading platform that continuously scans the market, analyzes hundreds of coins, identifies high-probability opportunities, manages risk, and executes trades automatically through supported exchange APIs.

2. Goals

Primary goal: trade only high-quality setups. Secondary goals include reducing emotional trading, minimizing unnecessary losses, automation, scalability, and long-term profitability.

3. Core Architecture

User Dashboard -> Authentication -> Portfolio Manager -> Trading Engine -> AI Decision Engine, Strategy Engine, Risk Engine, Indicator Engine, Market Scanner, News Engine, Whale Monitor, Order Manager -> Exchange APIs.

4. Major Features

Authentication, User Dashboard, API Key Management, Portfolio Analytics, Live P&L, Market Scanner, AI Scoring, Auto Trading, Risk Management, Notifications, Reports, Backtesting, Paper Trading, Multi-user SaaS.

5. Market Scanner

Scan 500+ coins, volume, liquidity, volatility, funding rate, open interest, BTC correlation, trend detection, breakout detection, and filter poor opportunities.

6. Technical Analysis

EMA, SMA, VWAP, SuperTrend, Ichimoku, RSI, MACD, ADX, ATR, Bollinger Bands, Stochastic RSI, OBV, MFI, Support/Resistance, Trendlines, Candlestick Patterns.

7. AI Decision Engine

Assign a confidence score using technical, volume, sentiment, volatility, market structure, and risk signals. Only execute trades above a configurable threshold.

8. Risk Management

Position sizing, daily loss limits, trailing stop, break-even, partial exits, exposure limits, cooldown after consecutive losses, emergency stop.

9. Strategies

Scalping, Swing, Momentum, Trend Following, Breakout, Mean Reversion, Grid, DCA, AI Adaptive Strategy.

10. News & Sentiment

Analyze news, Fear & Greed Index, social media, whale activity, exchange inflows/outflows, token unlocks, and listings.

11. Machine Learning

Learn from historical trades, optimize parameters, classify market regimes, and improve signal quality over time.

12. Tech Stack

Backend: Python, Django/FastAPI, Celery, Redis, PostgreSQL. Frontend: React/Next.js, TailwindCSS, TradingView Charts. AI: PyTorch, TensorFlow, Scikit-learn. Infrastructure: Docker, Kubernetes, Prometheus, Grafana.

13. Development Phases

Phase 1: Authentication, Exchange APIs, Dashboard. Phase 2: Indicators, Strategies, Auto Trading. Phase 3: AI Scoring & Risk Engine. Phase 4: ML & Analytics. Phase 5: SaaS & Multi-user.

14. Future Enhancements

Voice assistant, natural language trading, cross-exchange support, on-chain analytics, mobile apps, reinforcement learning research.

Important Note

No trading system can guarantee 90% profits or fixed maximum losses. The objective is to improve trade selection and automate disciplined execution using probability-based analysis.